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SEVENTH AND EIGHTH REVIEWS UNDER THE EXTENDED CREDIT FACILITY ARRANGEMENT AND REQUEST FOR WAIVERS OF NONOBSERVANCE OF PERFORMANCE CRITERIA—DEBT SUSTAINABILITY ANALYSIS

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Prepared by the International Monetary Fund and the International Development Association.

Ghana: Joint Bank-Fund Debt Sustainability Analysis	
Risk of external debt distress	*High*
Overall risk of debt distress	*High*
Granularity in the risk rating	*Sustainable*
Application of judgment	*No*

This debt sustainability analysis (DSA) concludes that Ghana's risks of external and overall debt distress continue to be assessed as high.[1] While the rebased nominal GDP significantly improved the public debt to-GDP ratios (albeit remaining elevated), the debt service ratios continue to breach their respective thresholds under the baseline, reflecting underlying vulnerabilities. The downward trend in total public debt-GDP ratio was interrupted in 2018, reflecting the realization of significant contingent liabilities in the banking sectors (seven banks resolved through end-December). At the same time, reflecting investor confidence in the first half of 2018 and efforts in restoring macroeconomic stability, Ghana successfully issued a US$2 billion Eurobond in May 2018 at more favorable terms relative to peer countries. Going forward, maintaining fiscal discipline, building buffers, diversifying the export base and exercising caution in contracting new external financing arrangements for infrastructure and other spending would be critical in making sure public debt dynamics be put firmly on a downward path.

[1] This DSA was jointly prepared by International Monetary Fund (IMF) and World Bank staff under the new debt sustainability framework (DSF) for low-income countries (LICs), implemented since July 2018. The LIC DSA compares the evolution of debt-burden indicators against thresholds, which are pre-determined depending on the classification of debt-carrying capacity (weak, medium, and strong). The debt-carrying capacity is classified using the country-specific composite indicator (CI) composed of three macroeconomic indicators and the World Bank's Country Policy and Institutional Assessment (CPIA). Ghana's capacity is assessed as "medium" using the CI based on the October 2018 WEO.

BACKGROUND

Public Debt Coverage

1. The DSA focuses on the public and publicly guaranteed (PPG) debt of the central government. In addition to the standard PPG debt, the DSA also captures debt contracted independently by selected energy-sector state-owned enterprise (SOEs), which are not explicitly backed by government guarantee and deemed to be the most important for the assessment of public debt dynamics.

	Subsectors of the public sector	Sub-sectors covered
1	Central government	X
2	State and local government	
3	Other elements in the general government	
4	o/w: Social security fund	
5	o/w: Extra budgetary funds (EBFs)	
6	Guarantees (to other entities in the public and private sector, including to SOEs)	X
7	Central bank (borrowed on behalf of the government)	X
8	Non-guaranteed SOE debt	X

2. Efforts to step up the government's capacity to record and monitor public debt and contingent liabilities are underway. The Ministry of Finance (MoF) prepared borrowing policy guidelines to regulate loan acquisition for ministries, departments and agencies (MDAs) and SOEs, which will be published once consultations with stakeholders are completed. In 2018, the government also launched a new credit risk assessment framework to enhance the management of fiscal risk arising from issuance of guarantees and on-lent operations to public entities as articulated in the Medium-Term Debt Strategy (MTDS) for 2019-22. Technical Assistance (TA) by the World Bank and IMF will continue in 2019 and include efforts to enhance data coverage of SOEs in the subsequent DSAs.[2]

3. The magnitude of the shock of the contingent liability test applied in the context of the sensitivity analysis of this DSA reflects potential additional liabilities. They could emanate from other

		Default	Used for the analysis	Reasons for deviations from the default settings
1	The country's coverage of public debt	The central government, central bank, government-guaranteed debt, non-guaranteed SOE debt		
2	Other elements of the general government not captured in 1.	0 percent of GDP	0	
3	SoE's debt (guaranteed and not guaranteed by the government) 1/	2 percent of GDP	2	
4	PPP	35 percent of PPP stock	1.44	
5	Financial market (the default value of 5 percent of GDP is the minimum value)	5 percent of GDP	5	
	Total (2+3+4+5) (in percent of GDP)		8.4	

1/ The default shock of 2% of GDP will be triggered for countries whose government-guaranteed debt is not fully captured under the country's public debt definition (1.). If it is already included in the government debt (1.) and risks associated with SoE's debt not guaranteed by the government is assessed to be negligible, a country team may reduce this to 0%.

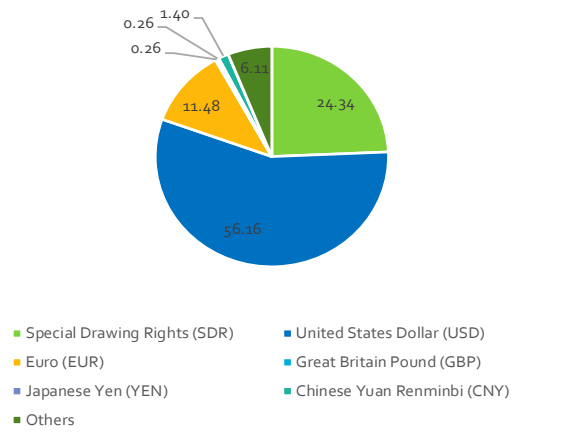
[2] World Bank TA in 2019 will be produced through the Debt Management Facility (DMF), the Ghana Economic Management Strengthening (GEMS) Project, and the Ghana Government Debt and Risk Management (GDRM) Project. TA through the DMF will be largely demand-driven. Other programmed project activities include: stakeholder engagement on the new borrowing guidelines and operational manual; Government staff training to prepare Annual Borrowing Plans in context of the MTDS and DSA, including sensitivity analysis and reporting; finalize the operational risk management (ORM) framework; support activities on reporting and communication; and develop capacity building on international capital market operations.

SOEs whose debts are not captured in the debt data coverage, public–private partnership agreement (PPP) and financial sector. Unanticipated bank resolution costs in 2018 and the 2017 issuance of the energy bond highlight the risk posed by contingent liabilities. Total contingent liabilities for the CL test are estimated at 8.4 percent of GDP.

Background on Debt

4. *The downward trend in total public debt-GDP ratio was reversed in 2018* due to financial sector cleaning up costs unanticipated during the last DSA. The fiscal performance faced headwinds in 2018 due to weak revenue collection (driven by lower VAT) and overruns in spending (driven by goods and services). Nonetheless, 2018 fiscal targets were still met thanks to combined effect of the July Mid-Year Budget Review measures, collection efforts in December and significant cuts in capital expenditures. However, including financial sector cleaning up costs (Staff Report Box 3), the primary deficit reached 1.4 percent of GDP compared a surplus of a 0.3 percent of GDP projected in the previous DSA.

Ghana: Currency Composition of External Debt End-July 2018



5. Ghana has increasingly relied on non-concessional borrowing along with its consistent improvement in income status and has maintained more-sustained market access. As of July 2018, Ghana's external public debt (excluding non-resident holdings of government securities) stood at US$18,211.8 or 28 percent of GDP. The borrowing comes mostly from multilateral creditors, bilateral creditors and international market. The main multilateral creditors are the African Development Bank (AfDB), International Development Agency (IDA) and IMF. Bilateral creditors are evenly spread across Paris and non-Paris club creditors. Over 50 percent of the external debt stock is denominated in US dollars.

UNDERLYING ASSUMPTIONS AND COUNTRY CLASSIFICATION

Background on Macro Forecasts

6. The updated macroeconomic projections assume a slightly lower short-to medium term growth trajectory compared to the April 2018 DSA and reflect the rebased nominal GDP, which increased 2017 nominal GDP by 25 percent (see Staff Report Box 1 table and box below).

7. While the authorities remain committed to contracting external loans in line with debt limits, other financing arrangements envisaged for 2019 pose potential risks to debt dynamics. They are within the 2018 debt limits for new non- concessional external borrowing priority projects set at US$3,500

million (cumulatively since end-Dec 2015). In 2018, the Parliament approved a financing arrangement with Sinohydro, a construction company, under which the latter will provide infrastructure worth US$2 billion to Ghana to be financed from Ghana's refined bauxite export proceeds (see Staff Report Box 5). The arrangement is planned to be independent of the government and to be implemented by a commercial entity established by the Parliament. While the arrangement—still to become effective—seems to have ring-fenced possible government obligations and therefore would not impact the DSA, it should still be designed and executed to ensure transparency and value for money. Since the project is classified as commercial debt, it is not included in the coverage of the DSA.

Ghana: Macroeconomic Assumptions Comparison Table

	DSA April 2018		Current DSA	
	2017-22	2023-2037	2018-23	2024-2038
Real GDP (%)	6.0	4.7	5.7	4.2
Inflation (GDP, deflator, %)	9.0	6.3	8.7	8.2
Nominal GDP (Billions of GHc)	301	1040	434	1814
Revenue and grants (% of GDP)	17.6	16.7	15.6	15.5
Grants (% GDP)	0.2	0.0	0.2	0.0
Primary expenditure (% GDP)	16.3	16.2	14.9	14.6
Primary balance (% GDP)	1.3	0.5	0.7	0.9
Exports of G&S (% change)	6.7	6.0	5.8	5.0
Noninterest current account deficit (% GDP)-	1.1	1.0	0.4	0.9

Sources: Ghanaian authorities and IMF staff estimates and projections

Box 1. Baseline Macroeconomic Assumptions

- **Overall real GDP growth** is estimated to have moderated to 5.6 percent in 2018 (1 percentage lower than the previous DSA estimate), reflecting lower growth in oil and gas than projected in the previous DSA. It is however projected to increase to 8.8 percent in 2019 on the back of increased oil production. Oil production is currently expected to peak in 2023, with the possibility of new oil discoveries and gas production implying significant upside potential. Non-oil growth reached 5 percent in 2018 and is expected to increase to 6 percent in 2019 and onwards, reflecting government initiatives to close infrastructure gaps and improvements in the financial sector. Tackling structural impediments including power supply, infrastructure, and diversifying the non-commodity economy will be key to increasing potential GDP.

- **Inflation and exchange rate:** After decreasing from 15.4 percent at end-2016 to 9.6 percent in April 2018, headline inflation has remained below, but stubbornly close to the 10 percent target upper-band, reaching 9.4 percent in December 2018. The end-period exchange rate came under pressure amid portfolio outflows, depreciating by over 8 percent at end-December 2018 relative to end- December 2017. Going forward inflation is expected to stay within the BOG's target band.

- **Government balance:** The 2019 budget targets an overall fiscal balance, excluding financial sector costs, of 4 percent of GDP and primary balance, excluding financial sector costs, of 1.6 percent of GDP. But high financial sector costs (not originally envisaged in 2019 under the program) will result in a primary balance of 0 percent of GDP. Yet, after these one-off costs, the primary balance is projected to remain around 1.5 percent of GDP over the medium term, provided additional domestic revenues are mobilized. Total public GFNs are projected decline by about 2 percent of GDP in 2019 but will remain elevated.

- **Current account balance:** The current account deficit narrowed to 3.2 percent of GDP in 2018 supported by oil exports. It is projected to further improve to 3 percent of GDP in 2019 reflecting sustained performance in the trade account supported by exports of key commodities. In the medium term, it would improve to an average of 3.2 percent of GDP; the non-interest current account will improve as well, reflecting maintenance of surpluses in the trade account. In line with the authorities' plans, 2019 will see an accumulation of foreign exchange reserves to restore buffers lost in 2018. Gross international reserves would steadily increase to almost 3-months of imports coverage by 2022.

- **Financing flows:** Mainly driven by the hydrocarbons sector, Ghana has enjoyed high FDI inflows in recent years, near 5½ percent of GDP in 2017. Looking forward, FDI is projected to decline gradually as oil production reaches its peak and eventually stays at around 4 percent of GDP over the long run. Consistent with Ghana's improving income status and more-sustained market access, grant inflows are projected to decline to around 0.1 percent of GDP in the medium term. Borrowing is projected to become increasingly non-concessional (less than a 35 percent grant element), as loans are expected to be used for key infrastructure projects to raise the potential growth rate. Up to US$3 billion Eurobond issuance is planned for 2019, of which US$1 billion is meant for liability management. A series of Eurobond issuances is envisaged to roll over maturing Eurobonds, which are assumed to be repaid on an amortizing basis rather than as bullet payments.

8. The realism tools show that projections are in line with historical and peers' experiences.

- **Fiscal adjustment**: The size of the projected fiscal adjustment over the three-year period up to 2020 is 1.1 percent of GDP (including financial sector cleaning up costs); and lies within normal ranges (below the top quartile of the distribution of past fiscal adjustments of LICs that have requested IMF supported programs).

- **Fiscal adjustment-growth**: In 2018, the difference between the expected fiscal impulse and the baseline can be explained by repairs and technology issues in key oil fields, which affected oil and gas production. However, in 2019, the baseline real GDP growth projection converges to the paths derived by using different typical multipliers in LICs.

- **Investment-growth:** The levels of private investment/GDP and public investment/GDP significantly shifted upward following the GDP rebase exercise, reflecting new data sources and a comprehensive survey. In terms of trends, private investment is expected to stay more or less flat throughout 2022 in line with the previous DSA but public investment will pick up starting in 2019 in line with the government policy of stepping up infrastructure projects. Consequently, the contribution of government investment to real growth is projected to be higher in the projection period than in the previous DSA.

- **Debt drivers**: for external PPG, debt dynamics in the projection period are mainly driven by current account deficit and FDI, the exchange rate, real GDP and nominal interest rates. Public debt dynamics will be driven by real growth and interest rates.

- **Forecast errors**: historical forecast errors for both external and public debt are largely driven by residuals, reflecting materialization of contingent liabilities. While the macroeconomic framework aims to capture such risks to the extent possible, the materialization of further contingent liabilities remains a relevant downside risk to the baseline.

Country Classification and Determination of Scenario Stress Tests:

9. Ghana is assessed to have a medium debt carrying capacity. Based on October 2018 WEO macroeconomic framework and the World Bank's CPIA index, Ghana's composite indicator score is 2.83 (above the lower cut-off value of 2.69 but below the strong capacity cut-off value of 3.05) confirming medium debt carrying capacity used in the April 2018 DSA under the old methodology. The new thresholds for a medium performer are therefore used below to assess external debt risk rating. Given Ghana's economic characteristics, tailored stress tests for commodity prices market financing, and contingent liabilities are applied.

Calculation of the CI Index

Components	Coefficients (A)	10-year average values (B)	CI Score components (A*B) = (C)	Contribution of components
CPIA	0.385	3.550	1.37	48%
Real growth rate (in percent)	2.719	5.490	0.15	5%
Import coverage of reserves (in percent)	4.052	24.294	0.98	35%
Import coverage of reserves^2 (in percent)	-3.990	5.902	-0.24	-8%
Remittances (in percent)	2.022	4.168	0.08	3%
World economic growth (in percent)	13.520	3.583	0.48	17%
CI Score			2.83	100%
CI rating			Medium	

(In percent)	Old methodology	New methodology
Public debt ratio-to-GDP	56	55
PV of PPG external debt-to-GDP	40	40
PV of PPG external debt-to-XGS	150	180
Debt service of PPG external debt-to-XGS	20	15
Debt service of PPG external debt-to-revenue	20	18

10. Contingent liabilities (CL) test is applied reflecting that SOEs and the financial sector represent a material risk to debt sustainability, as explained above. High levels of systemic losses and poor fee collection capacity in the utility sector could lead to fresh liabilities building up in the sector. While, the authorities have secured and set aside resources for the micro-finance sector resolution and are working on plans for resolution of the SDIs, risks from these sectors should be closely monitored.

EXTERNAL DSA

11. The external debt indicators have significantly improved relative to the previous DSA on account of the rebased GDP, but vulnerabilities associated with debt service remain (Figure 1 and Table 3). Two out of four indicators are in breach the thresholds under the baseline.[3] The debt service-to-exports ratio would be breached in the first 3 years; and then hovers around the threshold until 2022, breaches again from 2023 to 2026 beyond which it stays around the threshold. As in the past, debt service-to-revenue ratio is projected to stay above the threshold over the entire projection period. The outlook of this indicator would improve to the extent that the revenue mobilization measures are ramped up and through implementation of proactive debt management to further smoothen and lengthen the debt maturity profile.

12. The debt outlook remains sensitive to standard shocks under the DSA. The standard stress tests suggest that Ghana is particularly vulnerable to a decline in exports, confirming the need to diversify the economy and increase resilience to external shocks (Figure 1 and Table 3).

13. Ghana's gross financing needs have declined significantly over the past three years, but financing risks remain. This is flagged by the breach of benchmark for the market financing risk indicator of Ghana's GFNs, albeit with a small margin. The latest EMBI spread is just below the benchmark of 570, although there was a temporary breach between mid-November 2018 and end-January 2019 (Figure 5).

PUBLIC DSA

14. The public debt/GDP ratio improved following the GDP rebase. However, the downward trend in the debt/GDP was interrupted in 2018, reflecting the realization of significant contingent liabilities in the banking sector). Under the baseline, all debt indicators are expected to improve and stabilize reflecting the

[3] Under the new DSF, external debt indicators have been streamlined to only four indicators namely (i) PV of debt-to GDP ratio, (ii) PV of debt-to-exports ratio, (iii) Debt service-to-exports ratio and (iv) Debt service-to-revenue ratio.

expected fiscal consolidation and sustained growth. Nonetheless, total public debt-to-GDP ratio initially breaches the benchmark of 55 percent of GDP and would only decline below the benchmark by 2023 (Figure 2 and Table 4).[4] Over the medium term, debt service -to-revenue will also improve in the medium term.

15. Public debt sustainability is significantly vulnerable to commodity price shocks, with the PV of debt-to-GDP reaching 82 percent in 2028. On this basis, Ghana's overall public debt risk is assessed to be high.

CONCLUSIONS

16. The authorities agreed with the DSA results and are taking steps to improve debt dynamics and secure debt sustainability. Going forward, they have committed to cap overall fiscal deficit at 5 percent of GDP and secure positive primary balance, in line with the Fiscal Responsibility Law. In addition, they have set their own annual limits for the new external borrowing for projects and budget financing in 2019, following the same approach used under the ECF program.

17. The overall risk rating of debt distress is assessed to be high. The external risk of debt distress is assessed to be high because of the breaches of two indicators (external debt service-to-exports and external debt service-to-revenue) and the overall risk of debt distress is high because of those breaches as well as the PV of debt of total public debt-to-GDP ratio breaching its benchmark.

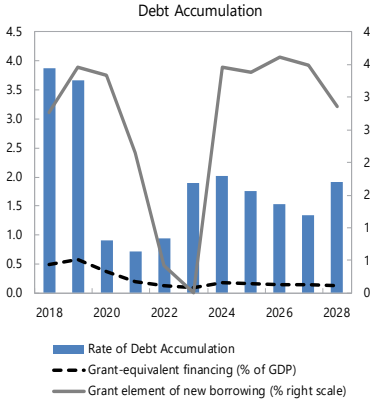
18. Ghana's public debt is deemed sustainable based on the authorities' sustained commitment to fiscal consolidation, prudent borrowing and proactive debt management initiatives, guided by their MTDS and recently passed Fiscal Responsibility Law. The average maturity of domestic public debt has been significantly increased in recent years and the recent buyback of the 2022 Eurobond with proceeds from the 2018 Eurobond helped to ease near-term refinancing risks. Going forward, keeping the fiscal performance on track is essential for engineering a favorable debt dynamic and maintaining confidence among foreign investors. In addition, efforts to strengthen the oversight over SOEs will be instrumental to stem their financial losses and prevent further feedback-loops to the budget and public debt. While Ghana's gross financing needs are declining, they will nonetheless remain high in the near term. Given the increasing presence of nonresidents in the domestic debt market, continued fiscal discipline and a strong macroeconomic environment are needed to continue attracting nonresident investors and anchor their confidence. A departure from the planned fiscal adjustment path could seriously jeopardize debt sustainability. Ghana should continue engaging with development partners to seek concessional financing and expedite disbursements under existing commitments from multilateral agencies.

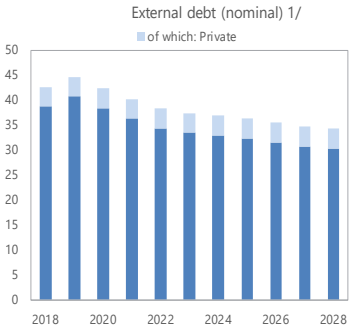
[4] As Ghana's debt carrying capacity is rated as medium, the relevant public debt benchmark is 55 percent of GDP.

Table 1. Ghana: External Debt Sustainability Framework, Baseline Scenario, 2015–38
(in percent of GDP, unless otherwise indicated)

	Actual			Projections								Average 8/	
	2015	2016	2017	2018	2019	2020	2021	2022	2023	2028	2038	Historical	Projections
External debt (nominal) 1/	**41.4**	**41.9**	**41.5**	**42.7**	**44.7**	**42.5**	**40.3**	**38.4**	**37.4**	**34.4**	**25.6**	**31.0**	**38.5**
of which: public and publicly guaranteed (PPG)	**37.4**	**37.9**	**37.5**	**38.7**	**40.7**	**38.5**	**36.3**	**34.4**	**33.4**	**30.4**	**21.6**	**26.9**	**34.5**
Change in external debt	9.0	0.5	-0.4	1.2	2.0	-2.2	-2.2	-1.9	-0.9	-0.3	-1.3		
Identified net debt-creating flows	**1.8**	**-6.1**	**-5.5**	**-3.4**	**-5.6**	**-3.2**	**-3.1**	**-3.4**	**-2.5**	**-1.1**	**0.1**	**-1.3**	**-2.5**
Non-interest current account deficit	**3.2**	**2.4**	**0.3**	**0.8**	**0.2**	**0.4**	**0.4**	**0.3**	**0.4**	**1.1**	**1.4**	**5.1**	**0.6**
Deficit in balance of goods and services	8.8	5.6	3.1	1.1	0.5	1.0	1.2	2.4	2.6	4.2	4.4	**8.8**	**2.5**
Exports	34.0	31.8	34.6	34.4	34.6	34.0	33.6	32.3	31.6	27.5	23.1		
Imports	42.7	37.3	37.7	35.5	35.1	35.0	34.8	34.7	34.2	31.7	27.4		
Net current transfers (negative = inflow)	-5.3	-2.6	-4.2	-4.0	-3.9	-3.7	-3.5	-4.2	-4.0	-3.1	-1.9	**-4.5**	**-3.7**
of which: official	-0.4	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Other current account flows (negative = net inflow)	-0.3	-0.5	1.4	3.6	3.5	3.1	2.7	2.1	1.8	0.0	-1.1	**0.9**	**1.8**
Net FDI (negative = inflow)	**-6.1**	**-6.3**	**-5.5**	**-4.5**	**-5.0**	**-4.2**	**-4.7**	**-4.9**	**-3.8**	**-2.7**	**-1.8**	**-6.0**	**-3.8**
Endogenous debt dynamics 2/	**4.7**	**-2.2**	**-0.3**	**0.3**	**-0.8**	**0.6**	**1.2**	**1.2**	**0.9**	**0.5**	**0.4**		
Contribution from nominal interest rate	2.6	2.8	3.1	2.4	2.8	3.0	3.0	2.8	2.6	1.9	1.5		
Contribution from real GDP growth	-0.8	-1.3	-3.2	-2.1	-3.6	-2.5	-1.8	-1.7	-1.8	-1.4	-1.1		
Contribution from price and exchange rate changes	2.9	-3.7	-0.2	...	...	...	...	...	...	...	...		
Residual 3/	**7.2**	**6.6**	**5.1**	**4.6**	**7.6**	**1.0**	**0.9**	**1.5**	**1.6**	**0.8**	**-1.3**	**3.0**	**1.8**
of which: exceptional financing	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Sustainability indicators													
PV of PPG external debt-to-GDP ratio	**...**	**...**	**36.8**	**36.8**	**38.7**	**37.4**	**35.5**	**33.8**	**33.0**	**30.1**	**22.1**		
PV of PPG external debt-to-exports ratio	**...**	**...**	**106.3**	**107.0**	**111.9**	**110.1**	**105.6**	**104.7**	**104.6**	**109.2**	**95.9**		
PPG debt service-to-exports ratio	**15.1**	**17.7**	**18.5**	**17.0**	**17.1**	**14.6**	**14.4**	**14.4**	**16.6**	**14.8**	**17.8**		
PPG debt service-to-revenue ratio	**38.4**	**43.7**	**47.9**	**41.1**	**37.3**	**32.9**	**30.7**	**29.5**	**33.0**	**26.0**	**27.1**		
Gross external financing need (Million of U.S. dollars)	1504.1	1443.0	1260.0	2074.6	1425.0	1584.6	1158.9	856.7	2610.5	4522.4	12228.9		
Key macroeconomic assumptions													
Real GDP growth (in percent)	2.2	3.4	8.1	5.6	8.8	5.8	4.4	4.5	5.0	4.3	4.4	**7.4**	**4.9**
GDP deflator in US dollar terms (change in percent)	-10.6	9.4	-0.8	4.6	-3.8	0.0	2.9	3.3	2.9	2.8	2.7	**-0.9**	**2.4**
Effective interest rate (percent) 4/	7.4	7.6	7.9	6.4	6.8	7.2	7.5	7.6	7.5	5.8	6.0	**4.5**	**6.9**
Growth of exports of G&S (US dollar terms, in percent)	8.1	5.9	17.0	9.8	5.1	4.0	6.3	3.5	5.7	5.3	3.6	**14.1**	**5.2**
Growth of imports of G&S (US dollar terms, in percent)	7.9	-1.1	8.4	4.1	3.4	5.6	6.7	7.5	6.5	6.0	3.8	**9.5**	**5.7**
Grant element of new public sector borrowing (in percent)	...	...	...	2.8	3.5	3.3	2.1	0.4	0.0	2.9	0.0	**...**	**2.6**
Government revenues (excluding grants, in percent of GDP)	13.4	12.9	13.3	14.3	15.8	15.1	15.8	15.8	15.9	15.6	15.1	**12.1**	**15.6**
Aid flows (in Million of US dollars) 5/	724.3	291.7	352.7	327.3	365.1	289.2	121.5	114.5	83.8	2.3	0.2		
Grant-equivalent financing (in percent of GDP) 6/	...	...	...	0.5	0.6	0.4	0.2	0.1	0.1	0.1	0.0	**...**	**0.2**
Grant-equivalent financing (in percent of external financing) 6/	...	...	...	6.1	7.3	9.7	5.9	3.6	1.9	2.9	0.0	**...**	**4.7**
Nominal GDP (Million of US dollars)	48,595	54,989	58,978	65,191	68,258	72,264	77,628	83,740	90,497	129,120	257,368		
Nominal dollar GDP growth	-8.6	13.2	7.3	10.5	4.7	5.9	7.4	7.9	8.1	7.2	7.2	**6.6**	**7.4**
Memorandum items:													
PV of external debt 7/	...	...	40.8	40.8	42.7	41.4	39.5	37.8	37.0	34.1	26.1		
In percent of exports	...	...	117.9	118.7	123.5	121.9	117.5	117.1	117.2	123.7	113.2		
Total external debt service-to-exports ratio	17.7	20.5	21.2	19.9	19.9	17.5	17.3	17.5	19.8	18.4	22.0		
PV of PPG external debt (in Million of US dollars)			21729.8	24018.3	26404.3	27029.2	27554.5	28286.6	29873.7	38830.5	56898.5		
(PVt-PVt-1)/GDPt-1 (in percent)				3.9	3.7	0.9	0.7	0.9	1.9	1.9	0.4		
Non-interest current account deficit that stabilizes debt ratio	-5.8	2.0	0.7	-0.4	-1.8	2.7	2.6	2.2	1.4	1.4	2.7		

Definition of external/domestic debt	Residency-based
Is there a material difference between the two criteria?	Yes





Sources: Country authorities; and staff estimates and projections.

1/ Includes both public and private sector external debt.

2/ Derived as [r - g - ρ(1+g) + Ɛα (1+r)]/(1+g+ρ+gρ) times previous period debt ratio, with r = nominal interest rate; g = real GDP growth rate, ρ = growth rate of GDP deflator in U.S. dollar terms, Ɛ=nominal appreciation of the local currency, and α= share of local currency-denominated external debt in total external debt.

3/ Includes exceptional financing (i.e., changes in arrears and debt relief); changes in gross foreign assets; and valuation adjustments. For projections also includes contribution from price and exchange rate changes.

4/ Current-year interest payments divided by previous period debt stock.

5/ Defined as grants, concessional loans, and debt relief.

6/ Grant-equivalent financing includes grants provided directly to the government and through new borrowing (difference between the face value and the PV of new debt).

7/ Assumes that PV of private sector debt is equivalent to its face value.

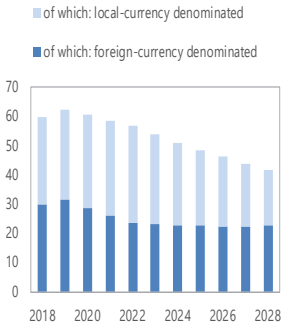
8/ Historical averages are generally derived over the past 10 years, subject to data availability, whereas projections averages are over the first year of projection and the next 10 years.

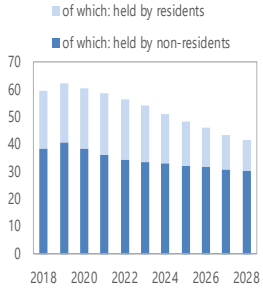
Table 2. Ghana: Public Sector Debt Sustainability Framework, Baseline Scenario, 2015–38
(in percent of GDP, unless otherwise indicated)

	Actual			Projections								Average 6/	
	2015	2016	2017	2018	2019	2020	2021	2022	2023	2028	2038	Historical	Projections
Public sector debt 1/	**55.2**	**57.5**	**57.7**	**59.8**	**62.4**	**60.4**	**58.6**	**56.7**	**54.0**	**41.5**	**26.2**	**42.2**	**53.0**
of which: external debt	**37.4**	**37.9**	**37.5**	**38.7**	**40.7**	**38.5**	**36.3**	**34.4**	**33.4**	**30.4**	**21.6**	**26.9**	**34.5**
Change in public sector debt	7.2	2.2	0.2	2.1	2.6	-1.9	-1.8	-1.9	-2.7	-2.2	-1.3		
Identified debt-creating flows	**3.7**	**2.4**	**-1.7**	**2.0**	**0.3**	**-2.5**	**-2.2**	**-1.9**	**-2.3**	**-2.0**	**-1.2**	**0.2**	**-1.6**
Primary deficit	**0.3**	**1.9**	**-0.5**	**1.4**	**0.0**	**-1.7**	**-1.6**	**-1.2**	**-1.3**	**-0.7**	**-0.8**	**2.4**	**-0.9**
Revenue and grants	14.9	13.4	13.9	14.6	16.1	15.3	15.9	15.9	16.0	15.6	15.1	**13.3**	**15.7**
of which: grants	1.5	0.5	0.6	0.3	0.3	0.2	0.1	0.1	0.1	0.0	0.0		
Primary (noninterest) expenditure	15.2	15.3	13.4	15.9	16.1	13.7	14.3	14.7	14.7	14.9	14.3	**15.7**	**14.8**
Automatic debt dynamics	**3.4**	**0.5**	**-1.1**	**-0.6**	**0.3**	**-0.8**	**-0.6**	**-0.6**	**-1.0**	**-1.2**	**-0.3**		
Contribution from interest rate/growth differential	1.9	1.5	-0.2	-1.0	-1.4	-0.8	-0.3	-0.4	-0.9	-1.1	-0.3		
of which: contribution from average real interest rate	2.9	3.3	4.1	2.1	3.4	2.7	2.2	2.1	1.9	0.7	0.9		
of which: contribution from real GDP growth	-1.0	-1.8	-4.3	-3.1	-4.8	-3.4	-2.6	-2.5	-2.7	-1.8	-1.2		
Contribution from real exchange rate depreciation	1.5	-1.0	-0.9	...	...	...	...	...	...	...	...		
Other identified debt-creating flows	**0.0**	**0.0**	**0.0**	**1.2**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**-0.1**	**0.1**
Privatization receipts (negative)	0.0	0.0	0.0	-0.5	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Recognition of contingent liabilities (e.g., bank recapitalization)	0.0	0.0	0.0	1.8	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Debt relief (HIPC and other)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Other debt creating or reducing flow (please specify)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Residual	**3.5**	**-0.1**	**1.9**	**0.5**	**4.1**	**0.5**	**0.1**	**-0.3**	**-0.5**	**-0.3**	**-0.2**	**2.6**	**0.1**
Sustainability indicators													
PV of public debt-to-GDP ratio 2/	**...**	**...**	**57.6**	**59.8**	**62.4**	**60.5**	**58.8**	**57.0**	**54.5**	**42.0**	**27.1**		
PV of public debt-to-revenue and grants ratio	**...**	**...**	**413.0**	**411.0**	**387.0**	**394.9**	**369.6**	**358.4**	**340.1**	**268.8**	**179.1**		
Debt service-to-revenue and grants ratio 3/	**119.7**	**133.3**	**134.8**	**91.7**	**74.9**	**73.2**	**65.4**	**67.4**	**73.2**	**50.3**	**35.1**		
Gross financing need 4/	18.1	19.8	18.2	14.7	12.0	9.6	8.8	9.5	10.4	7.1	4.5		
Key macroeconomic and fiscal assumptions													
Real GDP growth (in percent)	2.2	3.4	8.1	5.6	8.8	5.8	4.4	4.5	5.0	4.3	4.4	**7.4**	**4.9**
Average nominal interest rate on external debt (in percent)	8.4	8.1	8.8	6.6	7.4	7.8	8.0	8.1	7.9	6.1	6.3	**4.6**	**7.3**
Average real interest rate on domestic debt (in percent)	5.1	5.1	9.0	3.4	7.0	3.3	2.3	2.2	1.9	-0.5	2.4	**-7.5**	**1.8**
Real exchange rate depreciation (in percent, + indicates depreciation)	5.7	-2.9	-2.9	...	...	...	...	...	...	...	...	**3.2**	**...**
Inflation rate (GDP deflator, in percent)	13.6	15.2	10.4	10.3	6.7	8.6	8.9	9.0	8.6	8.5	6.4	**14.9**	**8.9**
Growth of real primary spending (deflated by GDP deflator, in percent)	-4.7	4.6	-5.4	25.7	9.8	-10.1	9.0	7.4	5.5	6.3	3.6	**6.9**	**6.2**
Primary deficit that stabilizes the debt-to-GDP ratio 5/	-6.9	-0.3	-0.8	-0.7	-2.7	0.3	0.2	0.7	1.4	1.4	0.4	**-2.7**	**0.6**
PV of contingent liabilities (not included in public sector debt)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		

Definition of external/domestic debt	**Residency-based**
Is there a material difference between the two criteria?	**Yes**

Public sector debt 1/





Sources: Country authorities; and staff estimates and projections.

1/ Coverage of debt: The central government plus social security and extra budgetary funds, central bank, government-guaranteed debt. Definition of external debt is Residency-based.

2/ The underlying PV of external debt-to-GDP ratio under the public DSA differs from the external DSA with the size of differences depending on exchange rates projections.

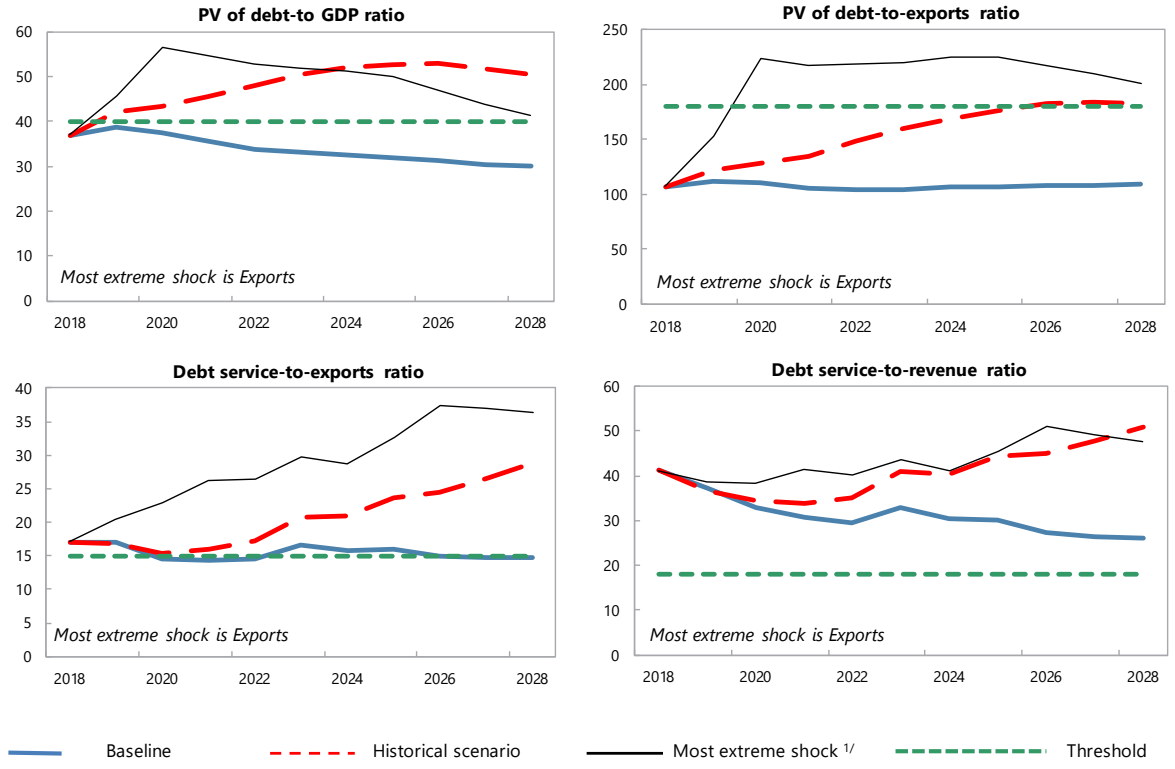
3/ Debt service is defined as the sum of interest and amortization of medium and long-term, and short-term debt.

4/ Gross financing need is defined as the primary deficit plus debt service plus the stock of short-term debt at the end of the last period and other debt creating/reducing flows.

5/ Defined as a primary deficit minus a change in the public debt-to-GDP ratio ((-): a primary surplus), which would stabilizes the debt ratio only in the year in question.

6/ Historical averages are generally derived over the past 10 years, subject to data availability, whereas projections averages are over the first year of projection and the next 10 years.

Figure 1. Ghana: Indicators of Public and Publicly Guaranteed External Debt Under Alternative Scenarios, 2018–28



Customization of Default Settings	Size	Interactions
Tailored Tests		
Combined CLs	No	
Natural Disasters	n.a.	n.a.
Commodity Prices 2/	No	No
Market Financing	No	No

Note: "Yes" indicates any change to the size or interactions of the default settings for the stress tests. "n.a." indicates that the stress test does not apply.

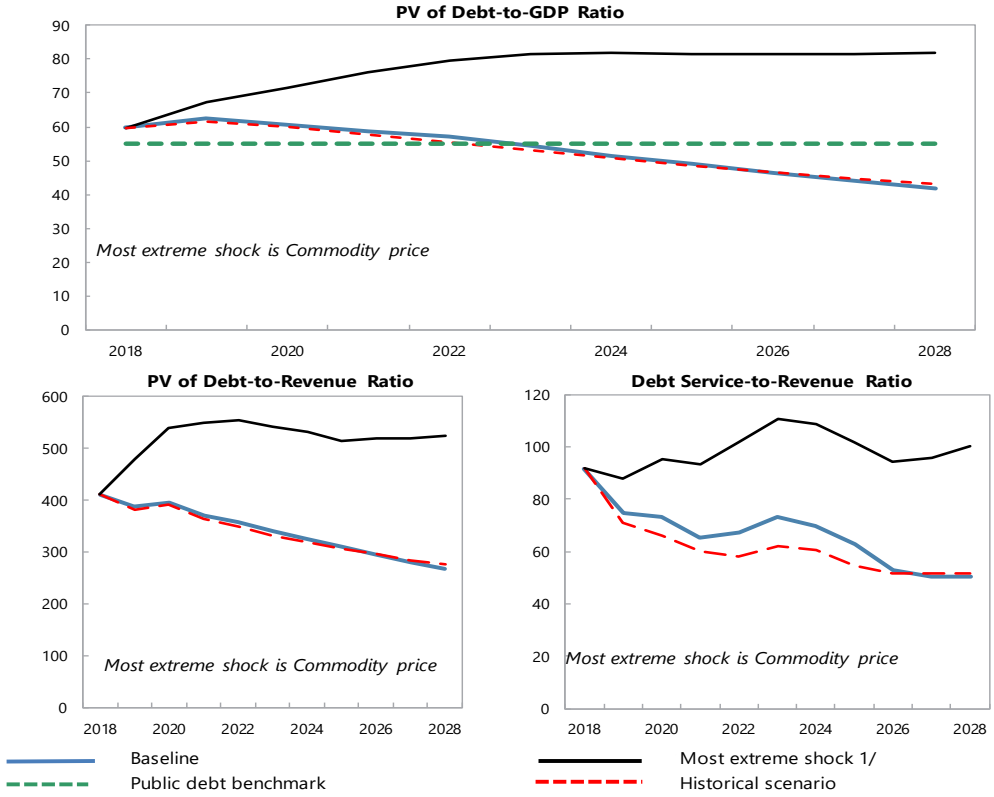
Borrowing Assumptions for Stress Tests*	Default	User defined
Shares of marginal debt		
External PPG MLT debt	100%	
Terms of marginal debt		
Avg. nominal interest rate on new borrowing in USD	8.9%	8.9%
USD Discount rate	5.0%	5.0%
Avg. maturity (incl. grace period)	10	10
Avg. grace period	5	5

* Note: All the additional financing needs generated by the shocks under the stress tests are assumed to be covered by PPG external MLT debt in the external DSA. Default terms of marginal debt are based on baseline 10-year projections.

Sources: Country authorities; and staff estimates and projections.

1/ The most extreme stress test is the test that yields the highest ratio in or before 2028. Stress tests with one-off breaches are also presented (if any), while these one-off breaches are deemed away for mechanical signals. When a stress test with a one-off breach happens to be the most exterme shock even after disregarding the one-off breach, only that stress test (with a one-off breach) would be presented.
2/ The magnitude of shocks used for the commodity price shock stress test are based on the commodity prices outlook prepared by the IMF research department.

Figure 2. Ghana: Indicators of Public Debt Under Alternative Scenarios, 2018–28



Borrowing Assumptions for Stress Tests*	Default	User defined
Shares of marginal debt		
External PPG medium and long-term	51%	51%
Domestic medium and long-term	35%	35%
Domestic short-term	15%	15%
Terms of marginal debt		
External MLT debt		
Avg. nominal interest rate on new borrowing in USD	8.9%	8.9%
Avg. maturity (incl. grace period)	10	10
Avg. grace period	5	5
Domestic MLT debt		
Avg. real interest rate on new borrowing	3.3%	3.3%
Avg. maturity (incl. grace period)	3	3
Avg. grace period	2	2
Domestic short-term debt		
Avg. real interest rate	2.3%	2.3%

* Note: The public DSA allows for domestic financing to cover the additional financing needs generated by the shocks under the stress tests in the public DSA. Default terms of marginal debt are based on baseline 10-year projections.

Sources: Country authorities; and staff estimates and projections.

1/ The most extreme stress test is the test that yields the highest ratio in or before 2028. The stress test with a one-off breach is also presented (if any), while the one-off breach is deemed away for mechanical signals. When a stress test with a one-off breach happens to be the most exterme shock even after disregarding the one-off breach, only that stress test (with a one-off breach) would be presented.

Table 3. Ghana: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt, 2018–28
(In percent)

	Projections 1/										
	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027	2028
PV of debt-to GDP ratio											
Baseline	37	39	37	35	34	33	33	32	31	30	30
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2018-2028 2/	37	**42**	**43**	**45**	**48**	**50**	**52**	**53**	**53**	**52**	**51**
B. Bound Tests											
B1. Real GDP growth	37	**42**	**43**	**41**	39	38	37	37	36	35	34
B2. Primary balance	37	**42**	**45**	**44**	**43**	**43**	**43**	**42**	**42**	**41**	**40**
B3. Exports	37	**46**	**57**	**55**	**53**	**52**	**51**	**50**	**47**	**44**	**41**
B4. Other flows 3/	37	**41**	**42**	40	38	37	37	36	35	33	32
B5. Depreciation	37	**47**	**43**	**41**	38	38	37	36	36	36	36
B6. Combination of B1-B5	37	**46**	**46**	**44**	**42**	**41**	**41**	**40**	39	37	36
C. Tailored Tests											
C1. Combined contingent liabilities	37	**44**	**43**	**41**	**41**	**40**	**40**	40	39	39	39
C2. Natural disaster	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C3. Commodity price	37	39	39	37	36	35	35	34	33	31	31
C4. Market Financing	37	**43**	**42**	**40**	38	38	37	36	35	34	33
Threshold	40	40	40	40	40	40	40	40	40	40	40
PV of debt-to-exports ratio											
Baseline	107	112	110	106	105	105	106	107	108	108	109
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2018-2028 2/	107	122	128	135	149	159	169	177	**183**	**184**	**183**
B. Bound Tests											
B1. Real GDP growth	107	112	110	106	105	105	106	107	108	108	109
B2. Primary balance	107	120	133	131	133	136	140	142	144	145	146
B3. Exports	107	153	**224**	**218**	**219**	**220**	**225**	**225**	**218**	**209**	**201**
B4. Other flows 3/	107	118	123	118	118	118	120	121	120	118	117
B5. Depreciation	107	107	99	95	93	93	94	96	98	99	102
B6. Combination of B1-B5	107	134	124	142	143	143	146	147	146	144	143
C. Tailored Tests											
C1. Combined contingent liabilities	107	126	127	123	127	128	131	134	136	138	140
C2. Natural disaster	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C3. Commodity price	107	112	113	110	110	111	113	113	113	112	112
C4. Market Financing	107	112	111	107	106	107	108	108	107	107	107
Threshold	180	180	180	180	180	180	180	180	180	180	180
Debt service-to-exports ratio											
Baseline	17	17	15	14	14	17	16	16	15	15	15
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2018-2028 2/	17	17	15	16	17	21	21	24	24	27	29
B. Bound Tests											
B1. Real GDP growth	17	17	15	14	14	17	16	16	15	15	15
B2. Primary balance	17	17	16	17	17	19	19	20	21	21	21
B3. Exports	17	20	23	26	26	30	29	33	37	37	36
B4. Other flows 3/	17	17	15	15	16	18	17	18	18	18	18
B5. Depreciation	17	17	14	13	13	16	15	14	12	12	12
B6. Combination of B1-B5	17	19	19	19	19	22	21	23	22	22	22
C. Tailored Tests											
C1. Combined contingent liabilities	17	17	16	16	16	18	18	18	17	17	17
C2. Natural disaster	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C3. Commodity price	17	17	15	15	15	18	17	17	16	16	16
C4. Market Financing	17	17	15	15	15	23	22	18	15	14	15
Threshold	15	15	15	15	15	15	15	15	15	15	15
Debt service-to-revenue ratio											
Baseline	**41**	**37**	**33**	**31**	**30**	**33**	**30**	**30**	**27**	**26**	**26**
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2018-2028 2/	**41**	**37**	**34**	**34**	**35**	**41**	**40**	**44**	**45**	**48**	**51**
B. Bound Tests											
B1. Real GDP growth	**41**	**41**	**38**	**35**	**34**	**38**	**35**	**34**	**31**	**30**	**30**
B2. Primary balance	**41**	**38**	**35**	**35**	**34**	**38**	**37**	**38**	**38**	**38**	**38**
B3. Exports	**41**	**38**	**38**	**41**	**40**	**44**	**41**	**45**	**51**	**49**	**48**
B4. Other flows 3/	**41**	**37**	**34**	**33**	**32**	**35**	**33**	**34**	**33**	**32**	**31**
B5. Depreciation	**41**	**48**	**41**	**37**	**35**	**40**	**37**	**34**	**29**	**28**	**28**
B6. Combination of B1-B5	**41**	**40**	**39**	**37**	**36**	**40**	**37**	**40**	**37**	**36**	**36**
C. Tailored Tests											
C1. Combined contingent liabilities	**41**	**38**	**36**	**34**	**33**	**37**	**34**	**34**	**31**	**31**	**30**
C2. Natural disaster	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C3. Commodity price	**41**	**44**	**39**	**37**	**35**	**38**	**34**	**32**	**29**	**28**	**28**
C4. Market Financing	**41**	**37**	**34**	**32**	**31**	**46**	**43**	**33**	**28**	**26**	**26**
Threshold	18	18	18	18	18	18	18	18	18	18	18

Sources: Country authorities; and staff estimates and projections.
1/ A bold value indicates a breach of the threshold.
2/ Variables include real GDP growth, GDP deflator (in U.S. dollar terms), non-interest current account in percent of GDP, and non-debt creating flows.
3/ Includes official and private transfers and FDI.

Table 4. Ghana: Sensitivity Analysis for Key Indicators of Public Debt, 2018–28

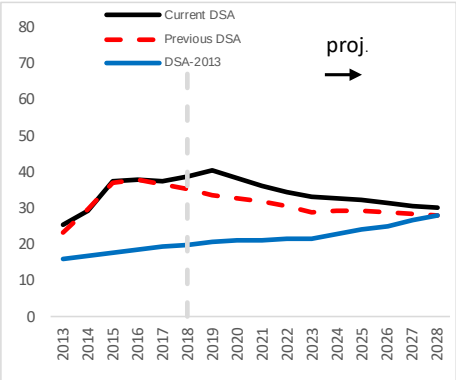
	Projections 1/										
	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027	2028
PV of Debt-to-GDP Ratio											
Baseline	**60**	**62**	**61**	**59**	**57**	55	52	49	47	44	42
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2018-2028 2/	**60**	**62**	**60**	**58**	**56**	53	51	49	47	45	43
B. Bound Tests											
B1. Real GDP growth	**60**	**70**	**74**	**74**	**75**	**74**	**74**	**73**	**73**	**73**	**73**
B2. Primary balance	**60**	**67**	**73**	**72**	**70**	**68**	**66**	**63**	**61**	**59**	**57**
B3. Exports	**60**	**68**	**77**	**75**	**73**	**71**	**68**	**64**	**60**	**55**	51
B4. Other flows 3/	**60**	**65**	**65**	**63**	**61**	**59**	**56**	53	50	47	44
B5. Depreciation	**60**	**68**	**65**	**63**	**60**	**58**	54	51	49	46	43
B6. Combination of B1-B5	**60**	**66**	**70**	**69**	**67**	**65**	**63**	**61**	**59**	**57**	55
C. Tailored Tests											
C1. Combined contingent liabilities	**60**	**71**	**70**	**68**	**67**	**65**	**62**	**60**	**58**	**55**	54
C2. Natural disaster	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C3. Commodity price	**60**	**67**	**72**	**76**	**80**	**81**	**82**	**82**	**82**	**81**	**82**
C4. Market Financing	**60**	**62**	**61**	**59**	**58**	**55**	52	49	46	44	41
Public debt benchmark	55	55	55	55	55	55	55	55	55	55	55
PV of Debt-to-Revenue Ratio											
Baseline	411	387	395	370	358	340	325	310	296	281	269
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2018-2028 2/	411	383	392	364	350	333	319	307	296	285	276
B. Bound Tests											
B1. Real GDP growth	411	434	479	465	468	463	463	462	464	464	467
B2. Primary balance	411	417	478	452	442	425	413	400	389	377	366
B3. Exports	411	424	502	472	460	441	426	406	380	352	326
B4. Other flows 3/	411	401	424	397	386	367	352	335	318	300	284
B5. Depreciation	411	421	426	395	380	359	341	324	309	292	278
B6. Combination of B1-B5	411	408	456	432	423	408	396	384	374	362	351
C. Tailored Tests											
C1. Combined contingent liabilities	411	442	454	429	419	403	390	377	366	354	343
C2. Natural disaster	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C3. Commodity price	411	479	540	550	555	543	532	515	518	520	523
C4. Market Financing	411	387	396	372	362	344	328	310	295	279	265
Debt Service-to-Revenue Ratio											
Baseline	92	75	73	65	67	73	70	63	53	50	50
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2018-2028 2/	92	71	66	60	58	62	60	55	52	52	52
B. Bound Tests											
B1. Real GDP growth	92	82	86	81	86	95	95	90	82	84	87
B2. Primary balance	92	76	81	82	84	94	85	79	73	71	72
B3. Exports	92	75	76	74	76	81	78	76	74	70	69
B4. Other flows 3/	92	75	74	68	70	75	72	67	58	56	55
B5. Depreciation	92	75	76	71	70	77	75	67	58	56	55
B6. Combination of B1-B5	92	75	77	78	76	89	82	73	65	61	61
C. Tailored Tests											
C1. Combined contingent liabilities	92	76	86	74	87	84	79	74	62	59	60
C2. Natural disaster	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C3. Commodity price	92	88	95	93	102	111	109	102	94	96	100
C4. Market Financing	92	75	74	67	69	86	83	66	54	50	50

Sources: Country authorities; and staff estimates and projections.
1/ A bold value indicates a breach of the benchmark.
2/ Variables include real GDP growth, GDP deflator and primary deficit in percent of GDP.
3/ Includes official and private transfers and FDI.

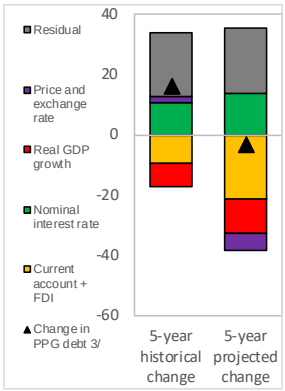
Figure 3. Ghana: Drivers of Debt Dynamics, Baseline Scenario

External Debt

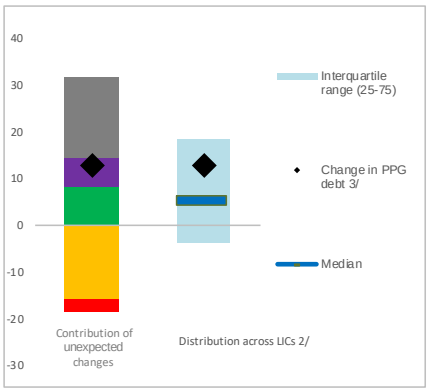
Gross Nominal PPG External Debt
(in percent of GDP; DSA vintages)



Debt-creating flows
(percent of GDP)

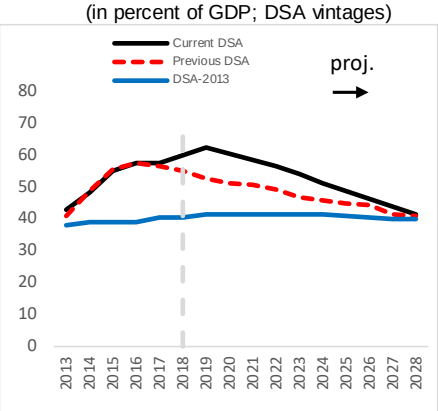


Unexpected Changes in Debt 1/
(past 5 years, percent of GDP)

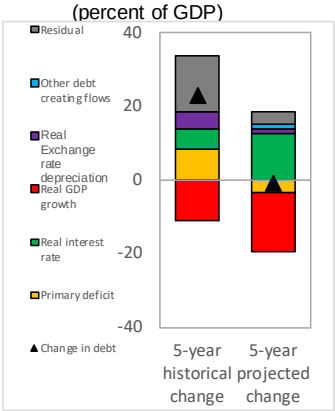


Public Debt

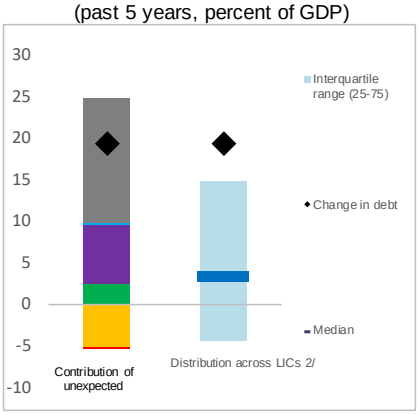
Gross Nominal Public Debt
(in percent of GDP; DSA vintages)



Debt-creating flows
(percent of GDP)



Unexpected Changes in Debt 1/
(past 5 years, percent of GDP)



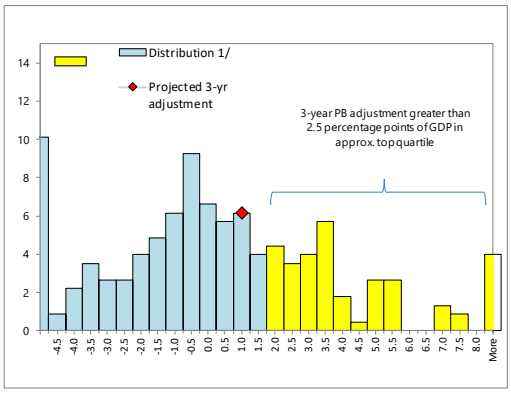
1/ Difference between anticipated and actual contributions on debt ratios.

2/ Distribution across LICs for which LIC DSAs were produced.

3/ Given the relatively low private external debt for average low-income countries, a ppt change in PPG external debt should be largely explained by the drivers of the external debt dynamics equation.

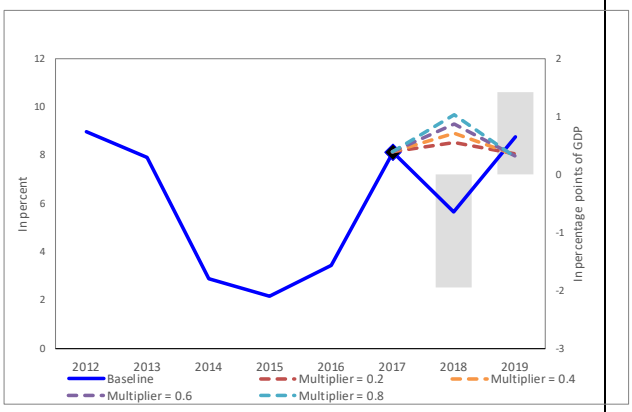
Figure 4. Ghana: Realism Tools

3-Year Adjustment in Primary Balance (Percentage points of GDP)



1/ Data cover Fund-supported programs for LICs (excluding emergency financing) approved since 1990. The size of 3-year adjustment from program inception is found on the horizontal axis; the percent of sample is found on the vertical axis.

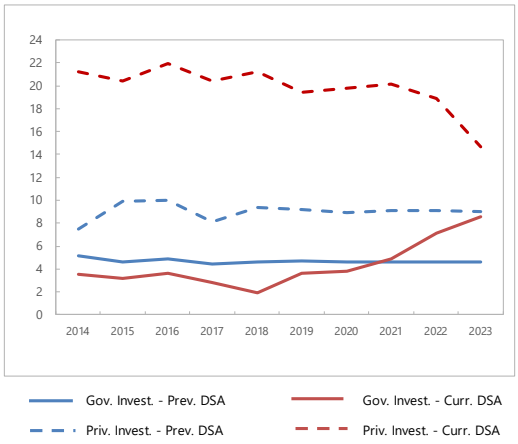
Fiscal Adjustment and Possible Growth Paths 1/



1/ Bars refer to annual projected fiscal adjustment (right-hand side scale) and lines show possible real GDP growth paths under different fiscal multipliers (left-hand side scale).

Public and Private Investment Rates

(percent of GDP)



Contribution to Real GDP growth

(percent, 5-year average)

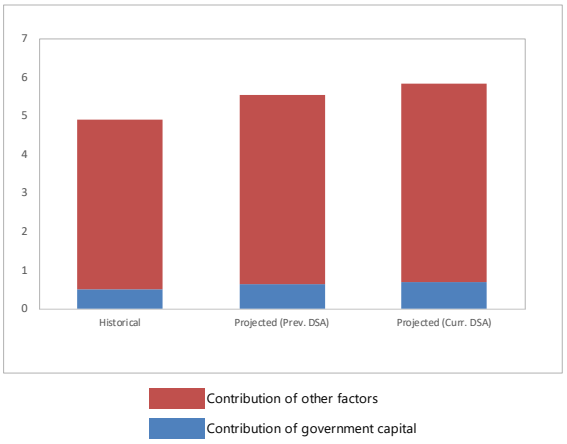
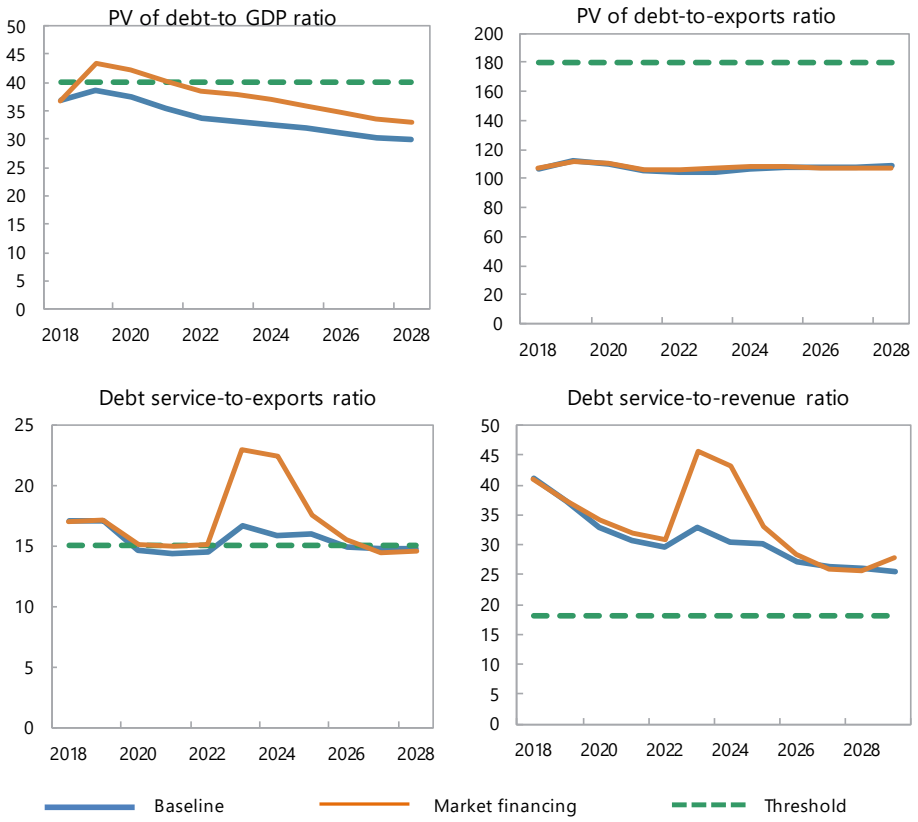


Figure 5. Ghana: Market-Risk Indicators

	GFN 1/	EMBI 2/
Benchmarks	14	570
Values	15	562
Breach of benchmark	Yes	No
Potential heightened liquidity needs	Moderate	

1/ Maximum gross financing needs (GFN) over 3-year baseline projection horizon.
2/ EMBI spreads correspond to the latest available data.



Sources: Country authorities; and staff estimates and projections.